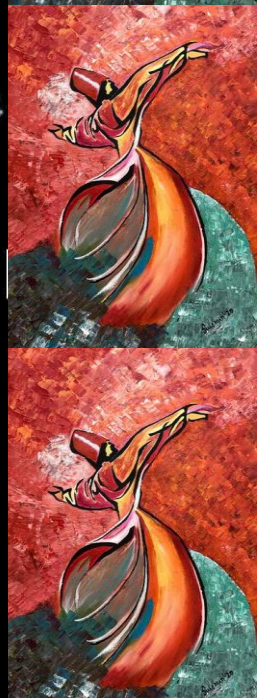




EE 354/CE 361/MATH 310 -
Introduction to Probability
and Statistics
Spring 2026



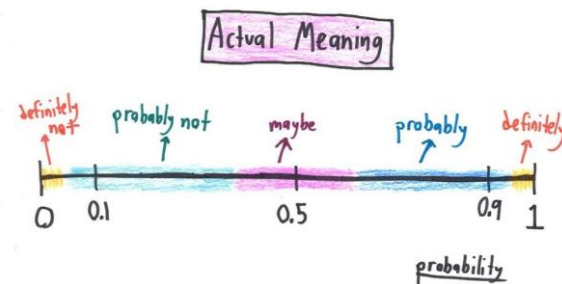


EE 354/CE 361/MATH 310 L4 section (Spring 2026)

Introduction to Probability and Statistics -

Aamir Hasan

Week 16 - Lecture No 30 – April 27, 2026



Announcements!

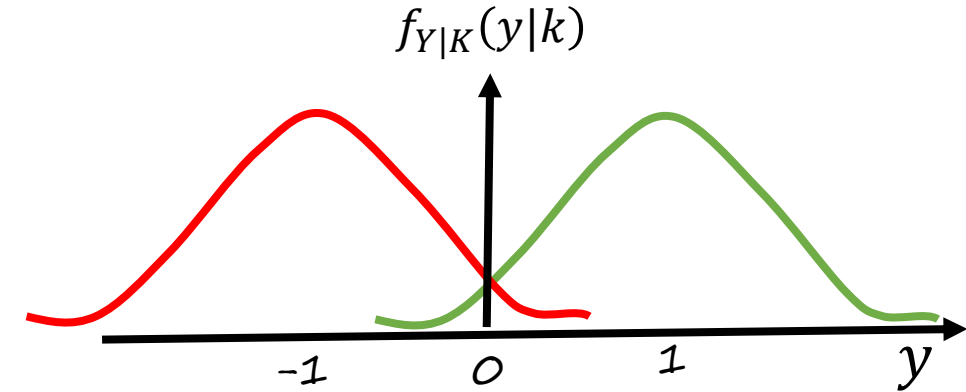
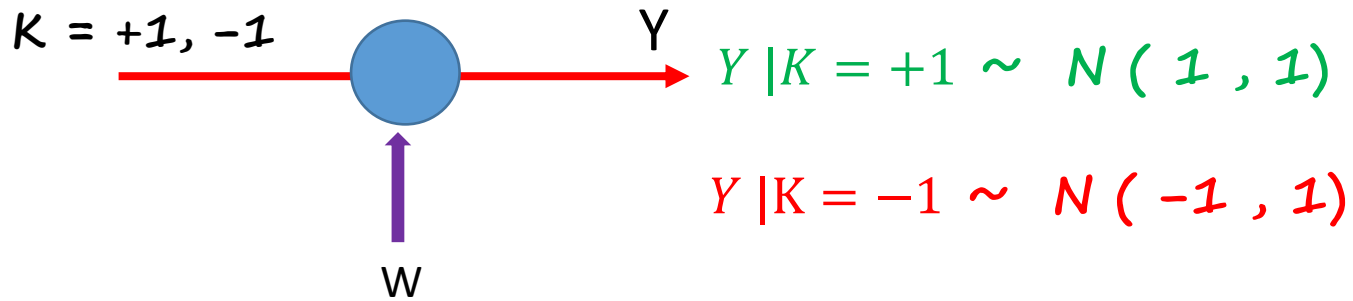
1. End-term Exam – Monday, May 4, 2026 (8:30a to 10:30a) @ Tariq Rafi, E-220
 1. Cheat Sheet – double-sided A-4 (HAND WRITTEN ONLY notes)
 2. Fall 2025 end-term uploaded
 3. 5 questions – 3 Continuous r.v.s, 1 discrete r.v.s and 1 from chapter 1
2. Planned quizzes
 - Quiz No 10 – April 29, 2026 (Wednesday) – Bayes Rule

Agenda for today

1. Unit 5 – Continuous Random Variable
 - ➡ ☐ Bayes Rule

The Bayes Rule – discrete unknown and continuous measurement

- Unknown K : equally likely to be $+1, -1$
- measurement Y : $Y = K + W$ $W \sim N(0, 1)$



- Probability that $K = +1$, given that $Y = y$? $P_{K|Y}(1|y)$

- $P_K(k) = 1/2$

$K = +1, -1$

- $f_{Y|K}(y|k) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(y-k)^2}$

K – Discrete, Y – Continuous (observing Y)

$$P_{K|Y}(k|y) = \frac{P_K(k) f_{Y|K}(y|k)}{f_Y(y)}$$

$$f_Y(y) = \sum_{k'} p_K(k') f_{Y|K}(y|k')$$

The Bayes Rule – discrete unknown and continuous measurement

- Unknown K : equally likely to be +1, -1

- measurement Y : $Y = K + W$

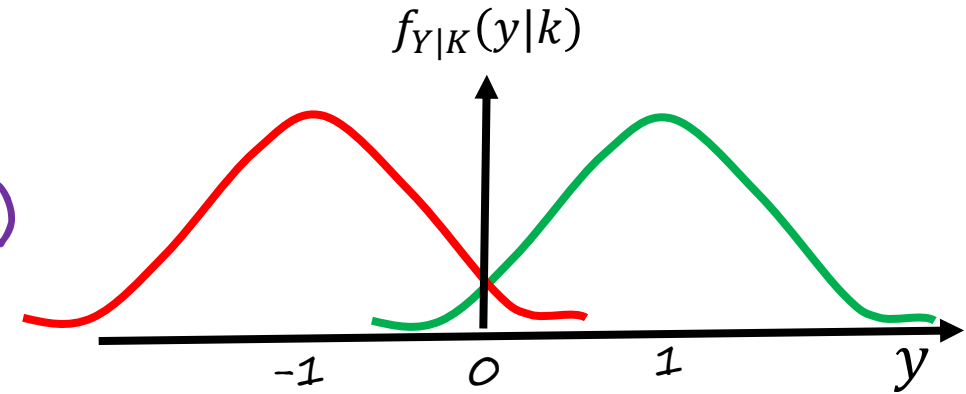
$$W \sim N(0, 1)$$

- ➡ • Probability that $K = 1$, given that $Y = y$?

$$f_Y(y) = \frac{1}{2\sqrt{2\pi}} e^{-\frac{1}{2}(y+1)^2} + \frac{1}{2\sqrt{2\pi}} e^{-\frac{1}{2}(y-1)^2}$$

$$P_{K|Y}(+1|y) = \frac{\frac{1}{2} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(y-1)^2}}{\frac{1}{2\sqrt{2\pi}} e^{-\frac{1}{2}(y+1)^2} + \frac{1}{2\sqrt{2\pi}} e^{-\frac{1}{2}(y-1)^2}}$$

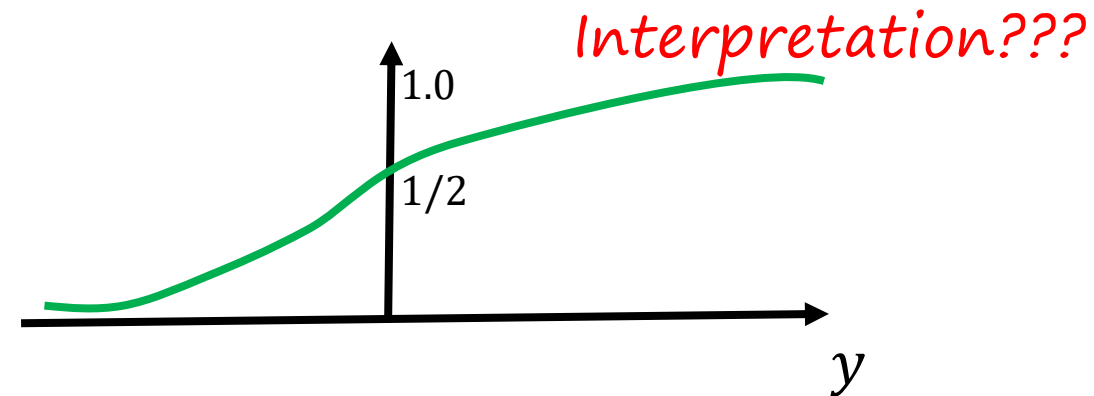
$$P_{K|Y}(+1|y) = \frac{1}{1 + e^{-2y}}$$



K – Discrete, Y – Continuous (observing Y)

$$P_{K|Y}(k|y) = \frac{P_K(k) f_{Y|K}(y|k)}{f_Y(y)}$$

$$f_Y(y) = \sum_{k'} p_K(k') f_{Y|K}(y|k')$$

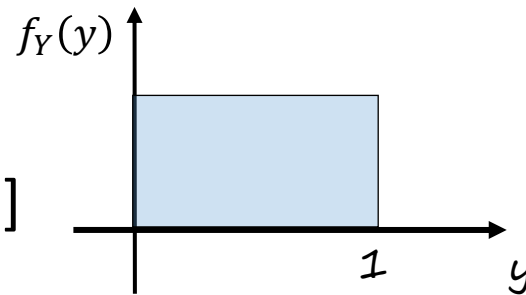
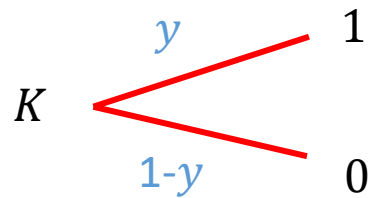




The Bayes Rule – *continuous unknown and discrete measurement*

The Bayes Rule – continuous unknown and discrete measurement

- K: Bernoulli with parameter Y



- unknown Y: uniform on $[0, 1]$

- Distribution of Y given that $K = 1$? $f_{Y|K}(y|1)$

$$f_Y(y) = \begin{cases} 1, & y \in [0, 1] \\ 0, & \text{otherwise} \end{cases}$$

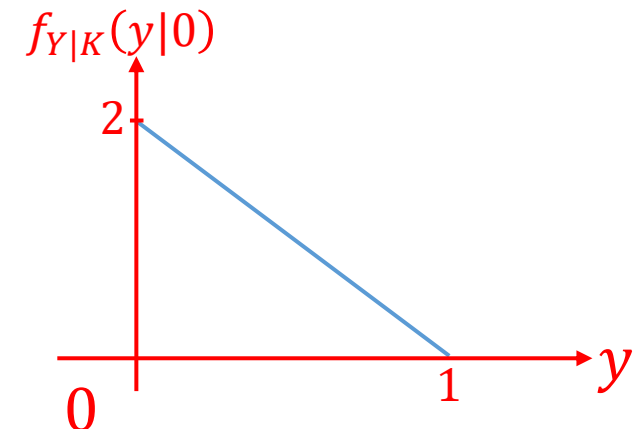
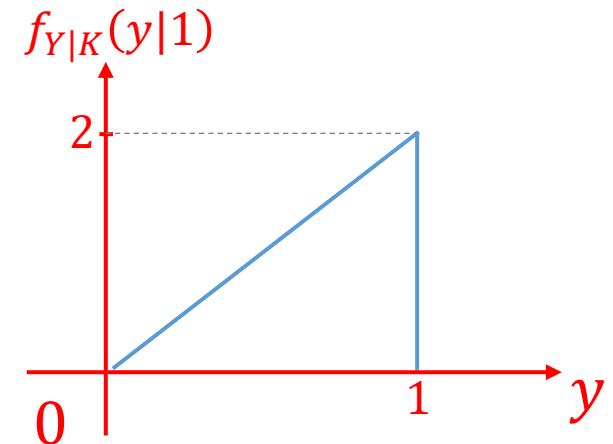
$$p_{K|Y}(1|y) = y$$

$$p_K(1) = \int_0^1 1 \cdot y \, dy = 1/2$$

$$f_{Y|K}(y|1) = \frac{1 \cdot y}{1/2} = 2y, \quad y \in [0, 1]$$

$$f_{Y|K}(y|k) = \frac{f_Y(y) P_{K|Y}(k|y)}{P_K(k)}$$

$$p_K(k) = \int f_Y(y') p_{K|Y}(k|y') \, dy'$$



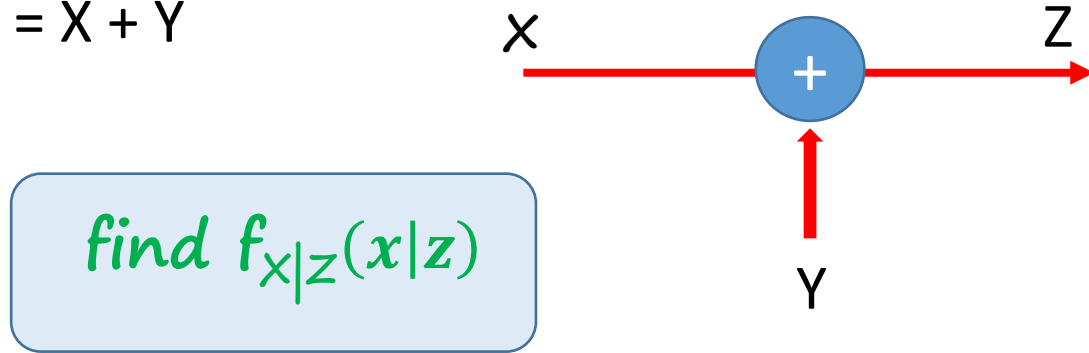


Example

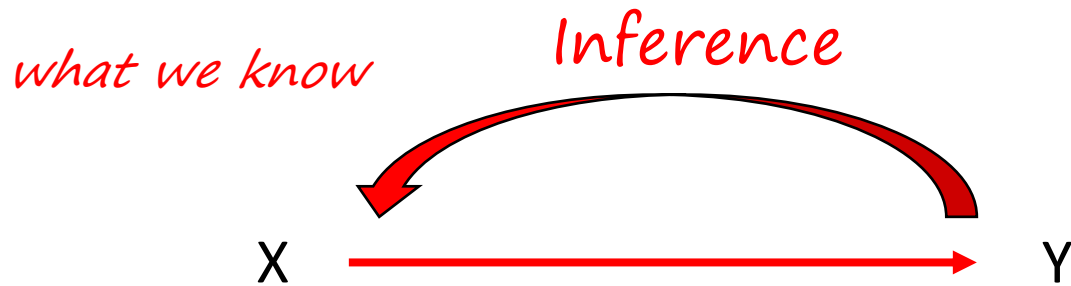
The Bayes Rule – continuous unknown and continuous measurement

The Bayes Rule – Example: Two continuous R.Vs

- X, Y independent continuous R.Vs (i.i.d)
- $Z = X + Y$



find $f_{X|Z}(x|z)$



- In this case, we observe Z and infer X

$$f_{X|Y}(x|y) = \frac{f_X(x)f_{Y|X}(y|x)}{f_Y(y)}$$



$$f_{X|Z}(x|z) = \frac{f_X(x)f_{Z|X}(z|x)}{f_Z(z)}$$

The Bayes Rule – Example: Two continuous R.Vs

- X, Y independent continuous R.Vs (i.i.d)
- $Z = X + Y$

find $f_{X|Z}(x|z)$

$$f_{X|Z}(x|z) = \frac{f_X(x)f_{Z|X}(z|x)}{f_Z(z)}$$

lets determine the model $f_{Z|X}(z|x)$ first

$$\text{we know } Z = X + Y \Rightarrow Z = x + Y$$

$$\Rightarrow Y = Z - x$$

$$\text{So, } f_{Z|X}(z|x) = f_Y(z-x)$$

Why?

$$\text{Since, } Z = x + Y$$

$$z = x + (z - x)$$

lets do it more formally!

The Bayes Rule – Example: Two continuous R.Vs

lets do it more formally!

$$\begin{aligned}P(Z \leq z \mid X=x) &= P(X+Y \leq z \mid X=x) \\&= P(x+Y \leq z \mid X=x) \\&= P(x+Y \leq z) \\&= P(Y \leq z - x)\end{aligned}$$

$$F_{Z|X}(z|x) = F_Y(z - x)$$

$$dF_Y(z - x)/dy = f_Y(z-x); \frac{dy}{dz} = 1$$

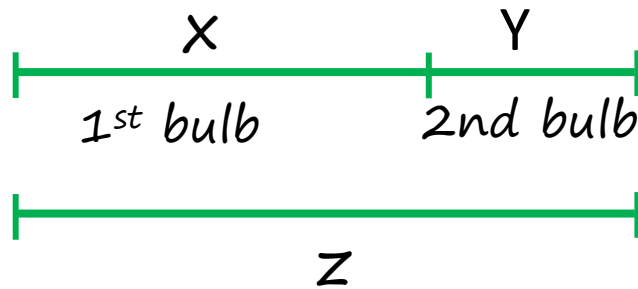
$$\text{so, } f_{Z|X}(z|x) = f_Y(z-x)$$

$$f_{X|Z}(x|z) = \frac{f_X(x)f_{Z|X}(z|x)}{f_Z(z)}$$

$$f_{Z|X}(z|x) = f_Y(z-x)$$

The Bayes Rule – Example: Two continuous R.Vs

- X, Y independent continuous R.Vs (i.i.d)
- $Z = X + Y$
- Now assume X, Y are exponential R.Vs with parameter λ (i.i.d)



$$f_{X|Z}(x|z) = \frac{f_X(x)f_{Z|X}(z|x)}{f_Z(z)}$$
$$f_{Z|X}(z|x) = f_Y(z-x)$$

You observe Z to make an inference about X (1st light bulb)

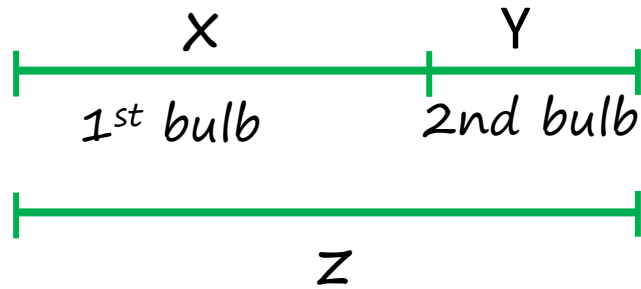
$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

$$f_Y(y) = \lambda e^{-\lambda y}, \quad y \geq 0$$

fix some $z \geq 0$

$$f_{X|Z}(x|z) = \frac{\lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)}}{f_Z(z)}$$
$$= \frac{1}{f_Z(z)} \lambda^2 e^{-\lambda z} \quad 0 \leq x \leq z$$

The Bayes Rule – Example: Two continuous R.Vs



for some z ; $f_Z(z)$ is a scalar quantity

$$f_{X|Z}(x|z) = \frac{1}{f_Z(z)} \lambda^2 e^{-\lambda z} \quad 0 \leq x \leq z$$

$$f_{X|Z}(x|z) = \text{uniform on } [0, z]$$

Why?

Think

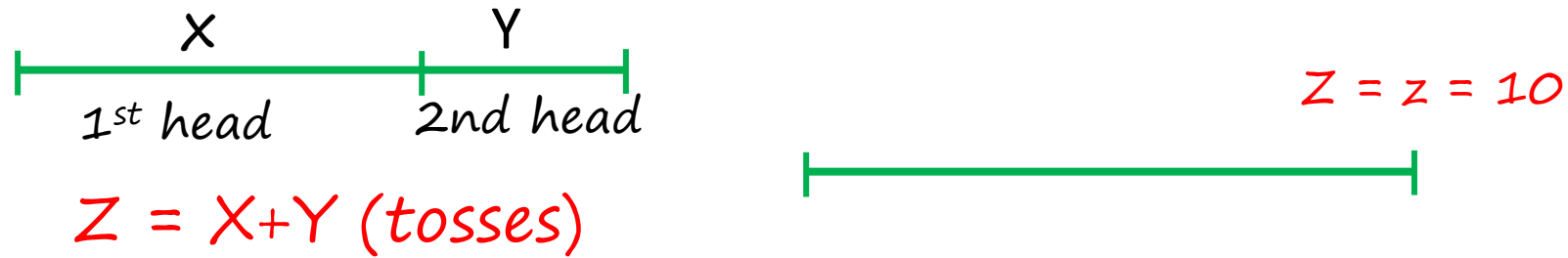


Discuss



Answer

The Bayes Rule – Example: Two independent tosses of coin



Given that 2nd head is on the tenth toss ($z = 10$), the first toss could be anywhere between 1 to 9

$$P(X=1|Z=10) = P(X=1, Y=9) = p \cdot (1-p)^8 p = p^2(1-p)^8$$

$$P(X=2|Z=10) = P(X=2, Y=8) = (1-p)p \cdot (1-p)^7 p = p^2(1-p)^8$$

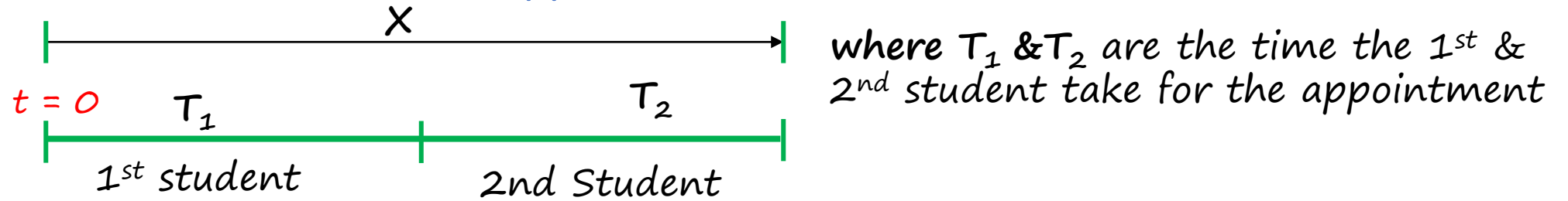
⋮

$$P(X=9|Z=10) = P(X=9, Y=1) = (1-p)^8 p \cdot p = p^2(1-p)^8$$

$P_{X|Z}(x|10)$ is uniformly distributed $X|Z \sim U[1, 9]$

Home Assignment

Example: Absent minded Professor – 2 appointments w/ two students at the same time



$$T_1 \text{ \& } T_2 \sim \exp(\lambda), \quad P(T_1 < t) = 1 - e^{-\lambda t}, \text{ for } t > 0 \quad P(T_2 < t) = 1 - e^{-\lambda t}, \text{ for } t > 0$$
$$E[T_1] = E[T_2] = 1/\lambda$$

Given that 2nd student arrives exactly after 5 minutes of the arrival of the 1st student

Comment on the $E[X]$ – where X is the total duration of the two appointments & for both students $\lambda = 1/30$ mins

Case 1 – condition $T_1 < 5$ mins $E[X \mid T_1 < 5] = 5 + E[T_2] = 35$ mins

Case 2 – condition $T_1 > 5$ mins $E[X \mid T_1 > 5] = 5 + E[T_1] + E[T_2] = 5 + 30 + 30 = 65$ mins

$$E[X] = P(T_1 < t) \cdot E[X \mid T_1 < 5] + P(T_1 > t) \cdot E[X \mid T_1 > 5]$$

Example Bayes Rule – Continuous – Discrete RVs

Example pertains to inferring a continuous random variable Q , from a discrete observation of a random variable X . Let Q be a continuous random variable with PDF as follows:

$$f_Q(q) = \begin{cases} 6q(1 - q), & \text{if } 0 \leq q \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Q represents the probability of a success of a Bernoulli random variable X , i.e., $P(X = 1 | Q = q) = q$ & $P(X = 0 | Q = q) = (1 - q)$. Find $f_{Q|X}(q|x)$ for $x \in \{0, 1\}$, for all possible values of q . You may use following version of Bayes' rule:

$$f_{Q|X}(q|x) = \frac{f_Q(q)p_{X|Q}(x|q)}{p_X(x)}$$

Given $f_{\theta|X}(u|x) = \frac{f_{\theta}(u) \cdot P_{X|\theta}(x|u)}{P_X(x)}$

where $P_X(x) = \int_0^1 f_{\theta}(u) \cdot P_{X|\theta}(x|u) du$ for $x=0,1$
for $x=0$ & $u \in [0,1]$

$$f_{\theta|X}(u|0) = \frac{6u(1-u) \cdot (1-u)}{\int_0^1 6u(1-u)(1-u) du} ; P_{X|\theta}(0|u) = (1-u)$$

$$= \frac{6u(1-u)(1-u)}{6 \int_0^1 u + u^3 - 2u^2 du}$$

$$= \frac{6u(1-u)(1-u)}{6 \left[\frac{u^2}{2} + \frac{u^4}{4} - 2\frac{u^3}{3} \right]_0^1} = \frac{6u(1-u)(1-u)}{\frac{1}{2}}$$

$$= 12u(1-u)^2$$

$$0 \leq u \leq 1$$

$$f_{Q|X}(q|x) = \frac{f_Q(q)p_{X|Q}(x|q)}{p_X(x)}$$

$$P(X=1|Q=q) = q \text{ \& } P(X=0|Q=q) = (1-q).$$

for $x=1$ and $q \in [0,1]$

$$f_{Q|X}(q|1) = \frac{6q(1-q) \cdot q}{1/2}$$

as $P_{X|Q}(1|q) = q$

$P_X(0) = 1/2 \Rightarrow$ ^{implies} $P_X(1) = 1/2$

$$f_{Q|X}(q|1) = 12q^2(1-q)$$

$$0 \leq q \leq 1$$

$$f_{Q|X}(q|x) = \frac{f_Q(q)p_{X|Q}(x|q)}{p_X(x)}$$

$$P(X=1|Q=q) = q \text{ \& } P(X=0|Q=q) = (1-q).$$

Congrats!

End of Unit 5!